To whom do women disclose Gender Based Violence (GBV) and why?

Kampe Ishimwe Rushoka

December 13, 2025

1 Introduction

Gender-based violence (GBV) remains one of the most widespread forms of violence, yet it also ranks among the least reported (Palermo et al., 2014). Victims worldwide face obstacles in disclosing their traumatic experiences to various institutions, whether the police, local authorities or healthcare professionals (Palermo et al., 2014). Institutions, whose administrative data collections is essential for national, regional, and international gender policy purposes (for Economic Co-operation & Development, 2018). These individual-level reporting challenges victims of GBV face, hence transcend into broader macro-governance challenges, as it undermine official GBV statistics to truly reflect the lived experiences of victims. Hence creating a reporting gap, that limits policymakers' ability to effectively design future GBV-responsive interventions (Richards et al., 2022).

This study aims to unpack the GBV reporting gap by first examining to whom women disclose GBV, and second, identifying the factors that shape these disclosure decisions. This trivial 21st century question is particularly urgent, due to the ongoing data revolution in public policy (Helbing, 2014). Now more than ever, diverse GBV data collection are essential to develop, implement and sustain strong gender responsive laws. (Schubotz et al., 2025).

While existing literature has examined reporting gaps between formal and informal avenues (Palermo et al., 2014), this paper argues that a state-affiliation analysis of reporting is necessary to account for the growing role of formal yet non-state institutions, such as women-focused non-governmental organizations (NGOs). Specifically, it compares reporting to state versus non-state affiliated institutions across the two epidemiological contexts of disclosure: self-initiated and solicited reporting (citations). Using a 2×2 typology of different GBV reporting contexts and scenarios, the study attempts to examines the rationale behind GBV victims' reporting decisions in each of these contexts.

These conceptual shifts aim to provide a nuanced understanding of the support systems that survivors rely on, and of whom they trust and turn to when navigating traumatic experiences such as domestic violence, rape, defilement, molestation, female genital mutilation, and other forms of gender-based violence.

2 GBV Reporting Data

Before examining reporting mechanisms on gender-based violence (GBV) it is first necessary to clarify its conceptual definition, as it directly shapes what can be captured in GBV data collection processes. Although GBV there have been scientific and cultural challenges in developing consistent indicators of GBV, the most widely adopted definition among researchers and policymakers has been the United Nations Declaration on the Elimination of Violence Against Women definition (Walby et al., 2017). GBV is defined as: "Any act of gender-based violence that results in, or is likely to result in, physical, sexual or mental harm or suffering to women, including threats of such acts, coercion, or arbitrary deprivation of liberty, whether occurring in public or private life" (United Nations General Assembly, 1993).

This definition has been academically preferred for it's broad and standardized nature, however different reporting studies have looked at more specific forms of GBV: including Intimate Partner Violence (IPV), Sexual Violence (SV), Sexual and Gender-Based Violence (SGBV), Female Genital Mutilation (FGM) and many more (Walby et al., 2017). Although the specific forms of GBV measured have varied across studies, this research deliberately adopts a broad, umbrella definition to focus on victims' reporting choices and decision-making in response to general gendered harm, rather than on distinguishing between specific types of harm they experienced. It is also important to acknowledge the limitations of the UN's definition, particularly as it focuses exclusively on women, overlooking the experiences of men and non-gender conforming individuals (Graaff, 2021). However, as extensive research suggest women and girls are globally the primary victims of gendered violence, this research considers it a reasonable assumption for our definition to focus on them (Umubyeyi et al., 2014).

2.0.1 Making Violence Visible : Methods of GBV Data Collection

GBV reporting and data collection is a robust science with various methods, and framings.

A prominent framing, is of GBV as a public health issue under the science of Epedimiology (Gracia, 2004). Epidemiology is known as the study of the distribution and prevalence of public health issues in larger populations (Edition, 2006). GBV prevalence has been historically justified as

a public health issue, as it can be measured, quantified, and analyzed at the population level. Moreover, it poses a significant public health threat due to its related consequences, including physical injury, mental health disorders, and reproductive health effects which are often addressed within public health systems. The scientific framing of GBV epidemiologically has allowed scholars and policymakers to scientifically record and estimate the prevalence and incidence of GBV all with the intent to reduce its occurence (Stöckl & Sorenson, 2024).

According to Epidemiological literature, public health reporting typically occurs through two main avenues active or passive surveillance[1]. Active surveillance involves health departments proactively contacting individuals of interest to solicit case information (solicited reporting), while passive surveillance relies on individuals voluntarily reporting notifiable diseases or adverse events to health institutions(self-initiated reporting) (Teutsch & Churchill, 2000). Although GBV has been widely recognized as an epidemiological issue, there is limited literature examining how GBV is differently reported in these two different contexts of self-initiated reporting and solicited reporting (Teutsch & Churchill, 2000).

Beyond its public health framing, GBV has also been framed as a social issue within academic research[2], a political issue addressed by feminist movements(particularly second-wave feminism) [3], and a security issue under the Women, Peace, and Security agenda of the United Nations Security Council [4]. While each framework has shaped GBV literature conceptually, the primary approaches used to estimate GBV prevalence data have been grounded in epidemiology and academic research, albeit employing a variety of data collection methods.

Ellsberg and Heise, in their influential article Researching Violence Against Women, broadly categorized GBV data collection into quantitative and qualitative approaches. Quantitative methods, which are predominantly used by epidemiologists, enable large-scale data collection to measure the prevalence and incidence of GBV and provide population-level insights. Such methods include cross-sectional surveys, cohort studies, and case-control studies. (Ellsberg et al., 2005; Stöckl &

[1]Epidemiologic surveillance is defined as the ongoing, systematic collection, analysis, and interpretation of health-related data, with the goal of preventing or controlling disease (Hall et al., 2012)

[2]Alldred, Pam. "Gender-Related Violence: What Can a Concept Do?" MDPI, 2023

[3]Khan, A. R. "The Role of the Second-Wave Feminist Movement and the Battered Women's Movement.", 2017

[4]United Nations Peacekeeping. "Women, Peace and Security." UN Peacekeeping, 2025

Sorenson, 2024). In contrast, qualitative methods, through interviews, ethnographies, and focus group discussions, focus on understanding the characteristics of lived experiences of GBV, within smaller or specific groups. Academics, independent researchers, and civil society organizations in the field of violence against women. primarily use qualitative methods to study violence against women (Ellsberg et al., 2005).

Although academic and public health experts have developed a variety of sophisticated methods for collecting and reporting GBV data, there is broad consensus that underreporting is widespread and that the data collected are far lower than true prevalence rates of violence experienced by women.

2.0.2 The Iceberg Effect in GBV Data Collection:

A well-known theory within the literature is the "GBV iceberg effect." In 2004, Enrique Gracia's widely cited feminist "Iceberg effect" theory claims that whatever understanding we gain of GBV through reporting represents only the visible "tip" of the problem (Gracia, 2004). He argues, the deeper, systemic magnitude of GBV remains largely hidden due to widespread stigma surrounding the issue. Gracia explains: "The 'iceberg' of domestic violence paints an image where reported cases of domestic violence against women are usually the most severe cases, and homicide of women by their intimate partners represents only the tip of the iceberg"(Gracia, 2004). According to this metaphor, most GBV cases remain submerged, invisible to society. Many scholars have expanded on this theory and even attempted to test it. Walby reinforced on the proposed iceberg effect; and argues that official statistics reflect only the extreme cases processed by state agencies, not the "real" rate of violence (Walby et al., 2017). Additionally, in 2013, Palermo, Bleck, and Peterman empirically evaluated disparities in GBV data collection, finding that estimates of GBV prevalence based on health system data or police reports significantly underestimate the true prevalence according to survey results of the DHS GBV module. They concluded that current methods for estimating GBV are misleading and fail to capture the full scope of the problem (Palermo et al., 2014). The Iceberg theory has immense policy implications. As it suggests that any GBV prevalence data, which informs public policy programming, funding, resource allocation, and awareness programs, is often drawn from the small subsample of individuals who decide to disclose victimization rather than represent the overall population of GBV victims (Palermo et al., 2014).

Beyond the iceberg theory, scholars have sought to explain GBV underreporting by examining the limitations of different reporting channels, particularly formal and informal channels.

2.0.3 (Under)Reporting of GBV: Limits, Violence, and Normalization in Formal and Informal Mechanisms

When examining GBV disclosure, the literature draws a sharp distinction between formal and informal GBV reporting mechanisms. What constitutes a "formal" source seems to historically vary across cultural and historical research, but according to Walby the common feature of formal GBV data collection is the production of systematically archived, documented accounts of GBV in institutions that have the potential of being analysed (Walby et al., 2017). Which include international and national-level surveys, smaller community-specific surveys, scale demographic, multi-year studies, official statistics, police records, recorded crime archives, public housing data etc (Walby et al., 2017). Walby argues that formal reporting mechanisms are useful for the breadth of information they have. Despite these strengths, other scholars have highlighted significant shortcomings in formal reporting mechanisms, particularly as victims don't deem them as "safe" or "reliable" avenues for reporting . In Improving Statistics on Violence against Women, Walby argues that police reports statistics systematically underestimate the prevalence of GBV as the majority of survivors choose not to report to the police. This behavior choice of the victim has been expanded by Ellsberg and Heise who argue that formal reporting mechanisms present multiple barriers to the survivor. Such as safety concerns, assault and revictimisation, fear of stigma, confidentiality issues, formal institutions failure to act and logistical obstacles such as the physical proximity to reporting centers (Ellsberg et al., 2005). Underscoring the incapacity of formal sources to fully capture GBV.

The relationship between formal institutions and GBV victims has been further explained by feminist theories on state violence. There is a feminist school of thought that views the state as fundamentally patriarchal. Drawing on structural violence theory, feminist Montesanti contend that social structures; including law, policy, and state institutions, produce and perpetuate gendered harm as patriarchal power relations are embedded in the very creation and functioning of the state (Montesanti, 2015). As second wave feminist Joan Acker notes, "gender is the foundation of hierarchical social relations, and these are deeply rooted in organizational and institutional

structures" (Acker, 1990). Underscoring how the exclusion of women from decision-making and citizenship rights has historically rendered the state patriarchal. These historical patterns, Montesanti argues, created conditions that make GBV both possible and persistent today (Montesanti, 2015).

Feminists Wilson, Ung Loh, and Purewal add to the conversation that, the persistence of high GBV prevalence is not merely a reflection of individuals violent actions but is evidence of broader structural and cultural acceptance of gendered violence within the state's institutions and practices (Wilson et al., 2018).

Focusing on specific state institutions, such as the criminal justice sector, the article "In an Abusive State", argues that courts responses to sexual and domestic violence often re-victimize survivors by prioritizing punitive power over survivor needs, funneling cases into legal systems that can re-traumatize women through skepticism, mandatory arrest policies, and adversarial court processes (Bumiller, 2008). Other feminist criminoligist highlight that police institutions have actively perpetuated harm against GBV survivors. Empirical research show that survivors' interactions with law enforcement can compound emotional trauma, as women are often disbelieved, dismissed, or treated with insensitivity by officers, resulting in psychological and emotional harm and discouraging future help-seeking (McQueen et al., 2021)

Collectively, feminist perspectives seem to strongly suggest that there are significant limitations to formal reporting, particularly to state institutions(such as the criminal justice system and the police). Especially as these structures have been both inactive in protecting survivors and have actively harmed them.

Informal GBV reporting, on the other hand, is less well defined in GBV scholarship. Across various studies, it has been considered as reporting to family, friends, neighbors, and religious institutions and is most often undocumented in nature (Walby et al., 2017). A study entitled "Formal and Informal Help-Seeking Associated with Women's and Men's Experiences of Intimate Partner Violence in Canada", found that informal sources of support, such as family and friends, were the

most common sources of support sought by women experiencing IPV. They found that women were more likely to speak with informal sources for emotional support and deemed those social networks safer and more reliable (Ansara & Hindin, 2010). This infamous study on GBV reporting also statistically showcased that informal reporting to social circle was vastly more common than formal reporting, particularly for Canadian regions more than others (Palermo et al., 2014).

However, scholars such as Enrique Gracia caution against labelling informal reporting as inherently more supportive and survivor friendly. He argues that though GBV might be invisible to the formal, through the iceberg effect, it might also be visible yet accepted in the informal. He suggests, many informal contexts and networks normalize GBV or even condone GBV reporting, making it equally inhospitable to survivors as formal settings (Gracia, 2004). Gracia highlights that survivors in highly violent or stigmatizing environments may turn to formal institutions not because formal institutions are inherently more protective, but because family, religious organizations, or local networks have normalized gendered violence and therefore fail to provide meaningful support. (Gracia, 2004). From this perspective, GBV reporting across formal and informal analysis reflects a paradox: in formal systems, violence may be rendered invisible through underreporting, while in informal systems, it may be visible but normalized, thereby undermining accountability.

2.0.4 NGOs: An Potential and Safer Avenue for GBV Reporting

There is growing literature on Non-governmental organisation (NGO)[5] GBV data collection, and their growing capacity and documenting GBV prevalence and instances. A case study from a women centered NGO in Kerashala India, highly demonstrates the potential of NGO-based reporting. The paper highlights Pechaan NGO's robust data collection on GBV instances, through their shelter archives, hotline records, crisis intervention analysis and their reports of community outreach(Kumari, 2023). Highlighting proximity of informal reporting survivors and their potential role in documenting GBV. Additionally, In the United Kingdom, a study drawing on records from UK women's shelters and refugee centers consistently documented more frequent abuse than formal

[5]NGOs are organizations that operate independently of any government, typically on a nonprofit basis, and are established to address social, cultural, environmental, or humanitarian issues (Willetts, 2002)

national statistics (Walby et al., 2017).

However, studies on NGO and social organisation GBV documentation remain rare, and there is limited scholarship on how it could be further leveraged systematically for GBV prevalence and incidence estimation. At the same time, there is an academic awareness on the limitations of Non-governmental sources in fully capturing GBV data. Ellsberg and Heise caution that NGOs specifically lack the scope and resources to implement large-scale national studies, which restricts their utility for global prevalence measurement despite their unique access to survivors (Ellsberg et al., 2005). Even less so for women's shelters, cooperatives, churches, mosques , and social networks, which may hold valuable information on GBV victims and their experiences but very rarely document it compared to NGOs.

3 Theoretical Framework

To address the question, "To whom do women report gender-based violence (GBV), and why?", this paper makes a twofold theoretical contribution to provide a nuanced understanding of the actors to whom GBV victims disclose.

The first theoretical contribution is a shift from a formality-based analysis of GBV reporting toward a state-affiliation framework. Traditionally, the literature has categorized GBV reporting as either formal (e.g., institutions) or informal (e.g., friends and family). However, this dichotomy overlooks the critical role of formally institutionalized yet non-state organizations, such as NGOs and community-based organizations. As highlighted earlier, many feminist scholars contend that state institutions are patriarchal in nature and often operate in ways that are hostile toward women, suggesting they may not be perceived as safe avenues for reporting. Emerging research underscores the potential for non-state affiliated institutions, such as NGOs, to be more survivor-centered and supportive (Kumari, 2023). Specifically, local women's NGOs and advocacy groups, frequently rooted in feminist principles and committed to advancing women's rights, could potentially be considered more survivor-friendly to victims, even though empirical evidence confirming this pattern remains limited (Kumari, 2023).

This paper's second theoretical contribution concerns examining GBV reporting within two epi-

demiological contexts: self-initiated reporting (when the GBV survivor voluntarily reports or seeks help) versus solicited reporting (when disclosure occurs in response to external questioning).

A substantial body of GBV methodology research demonstrates that who asks the question, and under what circumstances, dramatically influences whether survivors choose to reveal violence at all (Ellsberg et al., 2005). Yet, despite this clear evidence, no empirical work to date explicitly disentangles GBV data collected in the two contexts of self-initiated versus solicited disclosure within its analytical design. This absence is striking because the literature suggests that self-initiated disclosures often represent the most urgent or severe cases, reflecting situations where survivors exercise significant agency and capacity in seeking help (Ansara & Hindin, 2010). In contrast, solicited disclosures, those revealed through confidential surveys, capture a broader range of experiences, including those that remain hidden due to fear of retaliation, normalization, or shame (Ansara & Hindin, 2010).

Treating these two forms of disclosure as equivalent obscures the very mechanisms shaping help-seeking, institutional trust, and the lived realities of GBV; analyzing them separately is therefore not only methodologically necessary but theoretically essential. Hence, this epidemiological analysis could allows us to capture GBV reporting from both the highly motivated help-seekers and "silent survivors" whose experiences surface only in research contexts.

Hence, this paper presents a two-by-two typology, outlining four distinct GBV reporting contexts in Table 1, each illustrated with real-life examples of institutions that collect data within these respective contexts in Table 2.

Table 1: Two-Dimensional Typology of GBV Reporting

	State-affiliated	**Non-state-affiliated**
Self-initiated	Self-initiated & State-affiliated	Self-initiated & Non-state-affiliated
Solicited	Solicited & State-affiliated	Solicited & Non-state-affiliated

Table 2: Real life Manifestation of the Typology

Reporting Mechanism	**Situational examples**	**Institutional examples**
Self-Initiated and State Affiliated	Women reporting directly to the police, courts, or health professionals.	Example: FBI's *National Incident-Based Reporting System (NIBRS)* — the U.S. national database for all police-reported crimes, including GBV-related incidents. Captures offender details, demographics, and context of the offense (Federal Bureau of Investigation, 2023).
Solicited and State Affiliated	Women disclosing GBV through a national or governmental survey.	Example: The *National Crime Victimization Survey (NCVS)*—administered by the U.S. Census Bureau for the Bureau of Justice Statistics (BJS)—collects large-scale, nationally representative data on GBV experiences (Bureau of Justice Statistics, 2023).
Self-Initiated and Non State Affiliated	Women reporting GBV to NGOs, women's centers, hotlines, or shelters.	Example: Australia's *Women and Girls Emergency Centre (WAGEC)* collects hotline and crisis data on domestic and sexual violence. These data inform national statistics compiled by the Australian Institute of Health and Welfare (AIHW, 2022).
Solicited and Non State Affiliated	Women disclosing GBV in surveys conducted by NGOs or community-based organizations.	Example: The *National Sexual Violence Resource Center (NSVRC)* in the U.S. conducts large-scale community-based research to document GBV prevalence and promote prevention strategies (NSVRC, 2022).

Together, these tables demonstrate both the theoretical rationale and the empirical relevance of the proposed typology. Table 1 introduces the framework by categorizing disclosure along two dimensions: whether reporting is self-initiated or solicited, and whether the receiving institution is state-affiliated or non-state-affiliated. This structure moves beyond the traditional formal–informal distinction, emphasizing the role of state affiliation in shaping reporting pathways. Table 2 then illustrates how this typology operates in practice, providing real-life examples of institutions, such as police reporting systems, national surveys, NGO hotlines, shelters, and community-based research initiatives, that collect GBV data within each reporting context. Together, these tables underscore both the conceptual logic and the practical application of the paper's theoretical contributions,

situating GBV reporting within broader political and institutional frameworks.

This research employs the proposed typology to examine GBV disclosure and reporting across these distinct contexts. Based on this framework, I formulate two primary hypotheses with some subhypotheses:

H1: In the contexts of self initiated reporting, women are more likely to report Gender Based Violence a non-state affiliated institutions than a state affiliated institutions.

H1.1: Women with higher socioeconomic status are more likely to report GBV to state-affiliated mechanisms than to non-state mechanisms.

H1.2. Women with lower socioeconomic status are more likely to report GBV to non-state- affiliated mechanisms than state affiliated mechanisms.

H2: In the context of Solicited reporting, women are more likely to report (disclose) GBV to non-state institutions than state institutions.

These hypotheses are grounded in the feminist assumptions that non-state institutions are generally perceived as more survivor-friendly than state institutions to victims across both reporting context. The hypothesis also draw on economic evidence that socioeconomic factors, such as higher levels of education, urban residence, and greater access to services, shape individual's likelihood of interacting with state institutions such as the police, motivating the sub-hypotheses (Cattaneo & DeLoveh, 2008)

4 Methods

4.1 Methods for self-initiated reporting context

To test the first hypothesis, comparing reporting to state and non-state institutions, this study will draw primarily from the Demographic and Health Survey (DHS). It is a cross-sectional, nationally representative survey that collects standardized information across multiple countries that gather

data on socioeconomic characteristics, demographic indicators, and a range of health outcomes, allowing for cross-country comparisons and population-level analyses(Measure et al., 2012).

The survey includes a Gender-Based Violence (GBV) module, which ask 65 questions specifically related to GBV. It has been widely validated for reliability and comparability between countries, though minor variations exist across countries (Measure et al., 2012).

The GBV module targets a random sample of women of reproductive age (15–49 years), capturing both their experiences of violence from partner and family members as well as their reporting behaviors. Additionally, women are interviewed privately, independently of their partners, and their responses are kept confidential to ensure anonymity and safety(Measure et al., 2012).One limitation of this dataset is that it does not include younger girls or older women, who may also experience GBV.

The DHS GBV module includes questions directly relevant to this study, such as women's experiences GBV. Specifically, respondents are asked: "Has any previous husband/any other current or previous male partner/family member ever done any of the following things to you: ever emotionally abused you, ever pushed or hit you, ever thrown something at you, ever slapped you, ever kicked, dragged, or strangled you, ever threatened you with a knife or gun, ever physically forced you into unwanted sex, ever twisted your arm or pulled your hair, or ever committed other forms of physical violence (Measure et al., 2012).

The survey also collects information on victims' reporting behaviors and choices. Respondents are asked: "Thinking about what you yourself have experienced among the different things we have been talking about, have you ever tried to seek help?" They can indicate from whom they sought help: "from your own family, your husband's or partner's family, a current or former husband or partner, a current or former boyfriend, a friend, a neighbor, a religious leader, a doctor or other medical personnel, the police, a lawyer or criminal justice official, a local official, a social service organisation, a community organization" (Measure et al., 2012). This dataset therefore provides a representative snapshot of women's experiences with gender-based violence (GBV), including whether they sought help and, if so, from which type of institution.

This study focuses on DHS surveys conducted between 2005 and 2024 in India, Ghana, Nigeria, and Kenya, combining all survey waves conducted within that period. These countries were selected strategically because of prior civil society research that highlights strong presence of civil society, NGOs, and women's organizations, in all four. Providing an ideal context for examining and comparing GBV reporting behaviors across both state and non-state institutions (Akurugu et al., 2023). In contrast, countries with limited NGO activity would offer fewer opportunities for such comparisons, as reporting to non-state organizations would likely be rare or absent, limiting the study's ability to capture variation in reporting pathways.

4.1.1 Measures

To operationalize whether a respondent has experienced gender-based violence (GBV), a binary variable was constructed using the DHS dataset. Respondents were coded as 1 if they reported experiencing at least one instance of violence perpetrated by a current or former husband/partner, a close relative, or another individual within their social circle, as outlined by the DHS (Measure et al., 2012). This variable captures the broad GBV umbrella, including severe psychological, physical, and sexual violence within familial and social contexts, consistent with the United Nations' definition of violence against women (Walby et al., 2017). By incorporating multiple dimensions of violence, this measure ensures comprehensive identification of GBV experiences within the study population.

To capture self-initiated reporting, two sets of binary variables were created to distinguish between reporting to state-affiliated and non-state-affiliated institutions, in line with the study's theoretical framework. Respondents were coded as 1 for state-affiliated reporting if they reported seeking help from state institutions, including the police, public health clinic officials, or criminal justice officials; those who did not report to these institutions were coded as 0. For non-state-affiliated reporting, respondents were coded as 1 if they sought help from non-state institutions, including women's centers, social service organizations, religious institutions, community organizations or leaders, or family and friends, and 0 if they did not. This categorization follows the proposed theoretical framework, which distinguishes between state-affiliated and non-state-affiliated reporting mechanisms, allowing the analysis to capture survivors' preferences, perceived accessibility, and engagement with different institutional structures.

The final dataset for the four countries (India, Ghana, Nigeria, and Kenya) includes 362,588 observations of women who participated in the DHS Gender-Based Violence (GBV), and 126,614 who experienced GBV and 28,117 who reported GBV.

Figure 1 presents a barplot illustrating the distribution of the sample across countries and the prevalence of reported GBV. There is notable inter-country variation in both sample sizes and reported experiences of GBV. India has the largest sample, while Ghana has the smallest, reflecting the fact that Ghana conducted only two survey waves between 2005 and 2024, whereas the other three countries conducted three waves. Similarly, the proportion of respondents reporting GBV appears highest in India and lowest in Ghana, which is consistent with the relative differences in sample sizes across countries.

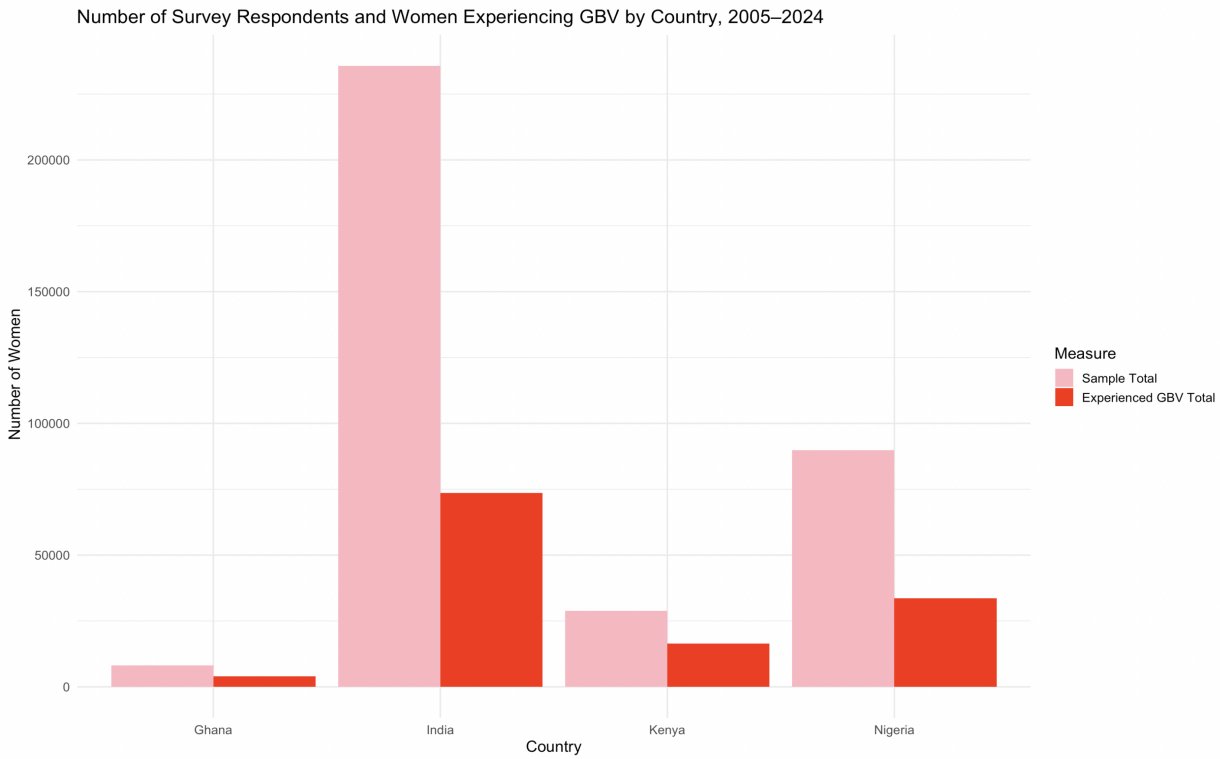


Figure 1: Side-by-side barplot showing total sample size and the number of women who experienced GBV in each country, based on DHS surveys from 2005 to 2024

.

4.1.2 Statistical Methods and Analysis

The statistical analysis for the first hypothesis was processed in two main steps, with sampled women aged 15–49 from India, Ghana, Nigeria and Kenya as the units of observation. First, preliminary descriptive statistics were used to compare the proportions of women who reported Gender-Based Violence (GBV) to state agencies versus non-state organizations, conditional on having both experienced and disclosed GBV.

Second, the study employed two binomial logistic regression models with Country fixed effects. The dependent variables are the two types of reporting: reporting to state institutions and reporting to non-state institutions, modeled separately in two independent regressions. These regressions are independent and different a women in the sample could report to both types of mechanisms, hence they are not mutually exclusive. The independent variables to both models are identical and they are the socioeconomic covariates, including age, educational attainment, wealth index, type of residence, number of children under 5, and country. A Logistic regression [6] is appropriate because the dependent variables are dichotomous, representing a binary reporting choice. Additionally, fixed effects [7] for country was included to account for cross-national differences in reporting behavior.

Mathematically, the models are be expressed as:

$$\begin{aligned}\text{StateReporting}_{i,c} = \beta_0 + \beta_1 \text{Age}_{i,c} + \beta_2 \text{Education}_{i,c} + \beta_3 \text{WealthIndex}_{i,c} + \beta_4 \text{ResidencyType}_{i,c} \\ + \beta_5 \text{NumChildrenUnder5}_{i,c} + \beta_6 \text{ExperiencedGBV}_{i,c} + \beta_7 \text{ReportedGBV}_{i,c} + \gamma_c + \epsilon_{i,c}\end{aligned} \tag{1}$$

$$\begin{aligned}\text{NonStateReporting}_{i,c} = \beta_0 + \beta_1 \text{Age}_{i,c} + \beta_2 \text{Education}_{i,c} + \beta_3 \text{WealthIndex}_{i,c} + \beta_4 \text{ResidencyType}_{i,c} \\ + \beta_5 \text{NumChildrenUnder5}_{i,c,t} + \beta_6 \text{ExperienceGBV}_{i,c} + \beta_7 \text{PriorReporting}_{i,c} + \gamma_c + \epsilon_{i,c}\end{aligned} \tag{2}$$

where:

[6] (LaValley, 2008)
[7] (Borenstein et al., 2010)

- i indexes individuals, c indexes countries, and t indexes survey year.
- $\text{StateReporting}_{i,c}$ and $\text{NonStateReporting}_{i,c}$ are binary indicators of whether a respondent reported GBV to a state- or non-state-affiliated institution, respectively.
- $\text{ExperiencedGBV}_{i,c}$ and $\text{ReportedGBV}_{i,c}$ are control variables capturing the necessary preconditions for reporting.
- γ_c are country fixed effects, accounting for time-invariant country-level factors.
- $\epsilon_{i,c}$ are errors due to unexplained residuals of the model.
- $\beta_0, \ldots, \beta_7$ are model coefficients, which, when exponentiated $(e^{\beta_0}, \ldots, e^{\beta_7})$, represent the change of odds of reporting to a state and a non-state institution.

The covariates in this study were measured as follows: age in years; highest education level categorized as no education, primary, secondary, or higher than secondary; wealth index score, a continuous measure derived from principal component analysis of household assets and living conditions; type of residence, coded as urban or rural; number of children under 5 years; and having experienced GBV and having reported, included as a control variable.

This model will help identify which individual-level factors are associated with a woman's likelihood of reporting GBV to state versus non-state mechanisms. Later, we will assess the model's inference by examining its standard errors and significance levels (p-values).

4.2 Methods for solicited reporting context

To test the second hypothesis, that in solicited contexts women are more likely to disclose experiences of GBV to non-state institutions than to state institutions, this research conducts a preliminary exploratory analysis comparing national GBV prevalence data from government surveys with estimates from NGO-led surveys. This analysis focuses on a case study of India, selected because it is one of the few countries where a large-scale GBV prevalence survey by a non-state organization was conducted in a similar geographic area and time frame as a state survey.

The NGO under consideration, the Society for Nutrition, Education and Health Action (SNEHA), is a Mumbai-based organization dedicated to advancing health equity among the urban poor. In

collaboration with the Institute for Global Health at University College London (UCL), SNEHA conducted a cross-sectional study titled Prevalence of Domestic Violence Against Women in Informal Settlements, Mumbai, 2017 (Daruwalla et al., 2020). India is also an ideal case because the National Family Health Survey (NFHS-4, 2015–16), conducted by the Ministry of Health and Family Welfare and the International Institute for Population Sciences (IIPS), collected GBV data in the same geographic area during a similar time period (**NFHS-4**). This overlap enables a meaningful comparison between state- and non-state-derived prevalence estimates.

This analysis aims to explore how NGO-reported GBV prevalence and explanatory narratives differ from national institutional statistics. Substantial differences, whether higher or lower, may indicate that victims are more likely to disclose GBV to non-state institutions than to state institutions.

5 Results

5.1 Results from Self-initiated Reporting

The descriptive statistics in Table 3 provide preliminary insights into the first hypothesis on self-initiated reporting, specifically comparing reporting to state versus non-state institutions accross countries. Table 3 presents the sample distribution, including the number and percentage of survey participants by country, the proportion who experienced GBV, and the proportion who reported it. The table further disaggregates reporting by state and non-state institutions.

Table 3: Descriptive statistics of DHS GBV module Survey Results by Country

	Ghana	India	Kenya	Nigeria
Measurement	n() | %	n() | %	n() | %	n() | %
Survey Participation	8179 | 2.26	235752 | 65.02	28901 | 7.97	89756 | 24.75
Base = country survey participation				
Experienced GBV	4055 | 49.58	73516 | 31.18	16349 | 56.57	33694 | 37.54
Base = having experienced GBV				
Reported GBV to someone	1292 | 31.86	13208 | 17.97	4868 | 29.78	8811 | 26.15
Reported To State	198 | 4.88	1189 | 1.62	622 | 3.80	1197 | 3.55
Reported To Non-State	193 | 4.8	913 | 1.2	827 | 5.1	1413 | 4.2

The results reveal notable variation across countries. Ghana, which had the smallest sample size, exhibits the highest overall reporting rate among women who experienced GBV, at 32 percent, while India, our largest sample country, has the lowest reporting rate of around 18 percent.

When examining reporting by institutional affiliation, interesting patterns emerge. First, the variation in reporting differs across countries. The direction of the comparison varies by country: Kenya (5.1% non-state vs. 3.8% state) and Nigeria (4.2% non-state vs. 3.6% state) align with Hypothesis 1, indicating that women are more likely to disclose to non-state-affiliated institutions than to state institutions. In contrast, Ghana and India show slightly higher state reporting than non-state reporting, though the differential margins are minimal (0.1% and 0.4%, respectively). Overall, the close reporting rates suggest the notion that women report to non-state and state-affiliated institutions at similar levels, and in some countries, may even report more frequently to non-state institutions.

Second, across all countries, both state and non-state reporting are significantly lower than general reporting. Suggesting that GBV victims often disclose to other actors before approaching institutions (either state or non-state) and highlighting the potential reporting to other mechanisms beyond the police, criminal justice sector, public health officials, social organizations, and community centers, as indicated by our initial model.

5.1.1 Inductive Results

Accounting for the missingness of a reporting actor(s) in Table 3, further investigation revealed that the majority of GBV disclosures did not occur through state or non-state institutions, but rather within social circles. Most reported instances were directed to family, friends, neighbors, strangers, or categorized as 'other,' reflecting social-circle reporting. Because our original categorization obscured these meaningful differences, we present a second descriptive statistics table (Table 4), structured similarly to Table 3, but further disaggregating reporting into three new distinct groups: State Institutions, Non-State Institutions, and Social Circle Reporting.

Table 4: Descriptive statistics summary by country, taking into account social circle reporting

	Ghana	India	Kenya	Nigeria
Measurement	n() — %	n() — %	n() — %	n() — %
Survey Participation	8179 | 2.3	235752 | 65	28901 | 8	89756 | 24.8
Base = country survey participation				
Experienced GBV	4055 | 49.6	73516 | 31.2	16349 | 56.6	33694 | 37.5
Base = having experienced GBV				
Reported GBV to someone	1292 | 31.9	13208 | 18	4868 | 29.8	8811 | 26.2
Reported To State	198 | 4.9	1189 | 1.6	622 | 3.8	1197 | 3.6
Reported To Non-State	193 | 4.8	913 | 1.2	827 | 5.1	1413 | 4.2
Reported To Social Circle	1223 | 30.2	12905 | 17.6	4429 | 27.1	8464 | 25.1

Table 4 presents similar proportions of reporting to state and non-state institutions, while additionally incorporating social circle reporting, in contrast to Table 3. It is evident that social circle reporting constitutes the overwhelming majority of GBV disclosures across all countries. In contrast, reporting to state and non-state institutions occurs at similar, much lower rates, suggesting that victims primarily turn to their social circles first, with roughly equal likelihood of choosing state or non-state institutions thereafter. Another notable finding from Table 4 is that the combined proportions of all 3 reporting mechanisms, exceed the total percentage of those who reported. This suggests that women often disclose to multiple institutions and networks. Across all countries, there is overlap in reporting, with the highest levels of overlap observed in Ghana, Nigeria, and Kenya (see Appendix Fig 3) This indicates that women in our sample utilized a combination of reporting institutions and mechanisms rather than relying on a single avenue.

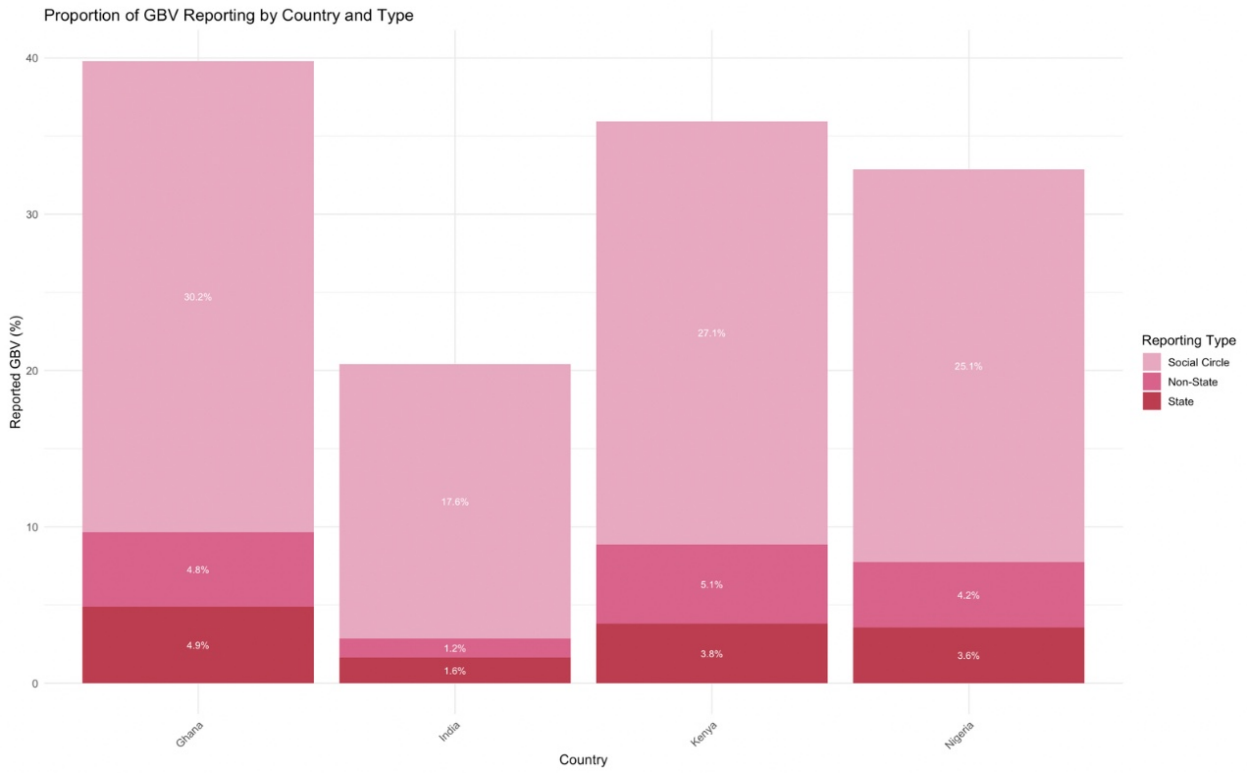


Figure 2: Stacked barplot showing reporting to social circle, state and non-state affiliated institutions according to country, based on DHS surveys from 2005 to 2024

The descriptive statistics are clearly displayed in Figure 2. The stacked bar graph illustrates reporting rates by country, with the height of each bar representing the total reporting rate per country, and the segments within each bar depicting the breakdown between social circle, non-state, and state reporting. Visually, we can see that Social circle reporting clearly dominates across countries, while non-state and state reporting are generally more comparable. In Kenya, non-state reporting is noticeably higher than state reporting, whereas in India, non-state reporting does not exceed state reporting. While these findings cannot fully support Hypothesis 1, that women, when self-initiating reporting, disclose more to non-state than to state institutions, the data show very comparable reporting rates, and in some countries such as Kenya and Nigeria, even higher reporting to non- state institutions. This suggests that women do utilize non-state institutions when reporting their traumatic experiences, and that data from such institutions, including NGOs, women's centers, community centers, and religious organizations, could make valuable contributions to broader GBV prevalence statistics, potentially even exceeding the reporting captured by state

institutions such as the police.

5.1.2 Factors Associated with State and Non-State GBV Reporting

To test the sub-hypotheses regarding socio-economic factors, the study conducted a fixed-effects logistic regression predicting reporting to state and non-state institutions. The results, presented in Table 5, reveal several significant relationships between women's characteristics and the type of institution they were associated with.

Table 5: Regression Results for State and Non-State GBV Reporting

Variable	Estimate	Std. Error	z value	p-value	Sig.
State Reporting (state_reporting = 1)					
Experienced GBV	-15.004	224.179	-0.067	0.947	
Respondent did not seek help from anyone	-35.952	257.409	-0.140	0.889	
Current age - respondent	-0.001	0.002	-0.542	0.588	
Educational attainment	0.022	0.004	4.984	0.000	***
Wealth index factor score (5 decimals)	0.000	0.000	0.203	0.838	
Type of place of residence (v025)	-0.196	0.041	-4.822	0.000	***
Number of children 5 and under in household (de jure)	-0.083	0.021	-4.014	0.000	***
Non State Reporting(non state reporting = 1)					
Experienced GBV	-9.439	325.031	-0.029	0.977	
Respondent did not seek help from anyone	-37.855	389.300	-0.097	0.923	
Current age - respondent	-0.022	0.005	-4.185	0.000	***
Educational attainment	-0.045	0.010	-4.283	0.000	***
Wealth index factor score (5 decimals)	0.000	0.000	0.129	0.898	
Type of place of residence (v025)	0.507	0.094	5.406	0.000	***
Number of children 5 and under in household (de jure)	0.149	0.053	2.814	0.005	**

Table 5 presents regression results in two panels, corresponding to models for state- and non-state reporting. The table displays coefficient estimates[8], standard errors, z-values, and p-values for each covariate.

For state-affiliated reporting (State reporting = 1), predictors of educational attainment, type of residence, and the number of children under five were statistically significant at a conventional margin of p ¡0.05.

While holding all other factors constant, a one-level increase in the educational attainment of a

[8]All coefficients are estimated in the log-odds scale. For interpretability, coefficients were exponentiated to represent odds ratios, which indicate the multiplicative change in the odds of reporting to an institution compared to not reporting, associated with a one-unit increase in the corresponding predictor.

GBV victim (e.g., from primary to secondary, or secondary to post-secondary) was associated with a 22% increase in the odds[9] of reporting their experience. Furthermore, GBV victims living in rural areas had 18% lower odds[10] of reporting to state institutions compared to if they lived in urban areas. Surprisingly, each additional child under five a GBV victim had was associated with an 8% decrease in the odds [11] of reporting to state institutions, controlling for all other covariates.

These results indicate that GBV victims reporting to state institutions are more likely to have higher educational attainment, live in urban areas, and have fewer children.

In contrast, the predictors of non-state reporting (Non-state reporting = 1) paint a different GBV victim. Age was interestingly is a significant predictor for non-state reporting though it wasn't for state reporting. Each additional year of age is associated with 2% lower odds[12] of reporting to non-state affiliated institutions. This means that, for women with the same background, a 50-year-old has approximately 60% higher odds of reporting to non-state institutions than a 20-year-old. This suggests that older women are more likely to report to non-state channels than younger women.

In contrast to state reporting, higher Educational attainment, residency type and number of children a woman had under the age of 5 had a negative effect effect on non-state reporting.

One level increase education was associated with 4% lower odds[13] of reporting to non-state institutions, suggesting that women with lower education levels are more associated with non-state institutions such as NGOs. Similarly, rural residency increased GBV victim odds of reporting with a 66% increase in odds [14] compared to if they lived in urban residency. Lastly, and each additional child under five was associated with a 16 increase in odds%[15] of reporting to non-state holding all other factors constant. Further contrasting with the patterns observed for state reporting.

These results indicate that GBV victims reporting to non-state institutions are more likely to have lower educational attainment, live in rural areas, and have more young children.

[9] $(e^{0.022} - 1 = 0.22)$
[10] $(e^{-0.196} - 1 = -0.18)$
[11] $(e^{-0.083} - 1 = -0.08)$
[12] $(e^{-0.022} - 1 = -0.02)$
[13] $(e^{-0.045} - 1 = -0.04)$
[14] $(e^{0.05} - 1 = 0.66)$
[15] $(e^{0.149} - 1 = 0.16)$

These results hint that distinct sociodemographic profiles shape how GBV victims turn to state and non-state institutions.

5.2 Results from solicited reporting

5.2.1 Exploratory Preliminary Hypothesis test on India

To assess whether, in solicited GBV disclosure, women report more frequently to non-state surveys than to state-led ones, this study conducted a preliminary comparison through the case study of India. Using SNEHA's NGO-led GBV survey report and the state-led National Family Health Survey summary report (NFHS-4) for the period 2016–2017 in Mumbai.

When comparing results, of the two reports, interesting findigs emerge. Both surveys estimated lifetime physical Intimate Partner Violence (IPV) at 21%, However all other estimates were different. Specifically their 12-month estimates of IPV diverged: SNEHA reported 15% of their Mumbai sample had experienced GBV in the last 12 months, whereas NFHS-4 found 11%. Notably, SNEHA's prevalence for sexual violence (5% lifetime; 4% past year) were higher than NFHS-4's corresponding estimates (2% lifetime; and 2% past year). For emotional violence IPV rates were likewise more elevated in SNEHA (17% lifetime; 12% past year) compared to the NFHS-4 (10% lifetime; 7% past year). Overall, for the urban Maharashtra sub-sample, SNEHA reported higher physical IPV: 21% in SNEHA vs. 16% in NFHS-4, sexual (5% in SNEHA vs. 2% in NFHS-4)and emotional violence (17% in SNEHA vs. 9% in NFHS-4). Across all measures, SNEHA's NGO-led survey consistently reported higher GBV prevalence than the state survey conducted by the Ministry of Health and Family Welfare (NFHS-4).

Several possible explanations account for these discrepancies. First, the survey instruments differed. SNEHA's dedicated GBV questionnaire included additional items on social control and emotional violence, leading to differences in the operationalization of GBV compared with the NFHS-4 domestic violence module (SNEHA & UCL, 2020, p. 5). Second, the sampling contexts were not completely identical. Although both targeted urban Mumbai, SNEHA surveyed women specifically in informal settlements, a population likely at higher risk than the broader urban sample included in NFHS-4. Third, consistent with our hypothesis, it is possible that non-state solicited measures

capture higher prevalence because of stronger survivor-centered support structures during data collection. SNEHA's final report stated that they implemented a "strong programme of support for survivors" and provided months-long training for interviewers on question framing and sensitive interviewing techniques, which might have encouraged greater disclosure (SNEHA & UCL, 2020, p. 9). This suggests that the conditions under which the SNEHA survey was conducted and survivor-friendly approach may have facilitated more honest reporting than the NFHS-4, where GBV questions appear as part of a broader health questionnaire.

However, this constitutes only a preliminary hypothesis test and cannot definitively determine whether women disclose more to non-state than to state-affiliated actors. Given that such a conclusion would require experimental testing with both survey types administered to the same population using identical instruments. These comparisons nonetheless suggest that non-state solicited surveys did elicit higher levels of GBV disclosure in the case study of India.

6 Discussion and Limitations

The results of this study provide several important insights into women's reporting behaviors in the context of gender-based violence (GBV) and highlights the critical yet overlooked role of non-state actors.

Across countries, self-initiated reporting showed that women reported to non-state actors at rates comparable to and in some countries such as Kenya and Nigeria higher than, reporting to state institutions. However, these patterns varied considerably across countries, suggesting that the social, institutional, and cultural setting shapes women's decisions about where to seek help. In the case of Kenya, this may be linked to the strong presence of feminist movements and women's rights organizations that actively receive GBV cases, provide legal aid, and advocate for survivors' rights, such as the Coalition on Violence Against Women (COVAW), which has worked since the mid-1990s to challenge harmful norms, expand equitable access to justice, and support gender equality in the capital Nairobi and in rural regions(Ali, 2017). Institutionally, it is plausible that COVAW is perceived as a more accessible and approachable reporting option than the police, particularly for women in rural areas and those with lower levels of education. Overall, these findings highlight the

need for further country-specific research that examines GBV reporting patterns across state and non-state institutions, as cross-national generalizations may obscure important local dynamics.

Inductively, we also found that most reporting did not occur through formal institutions but was directed toward women's social circles. Previous research has largely dichotomized reporting into formal (institutional) and informal (friends and family) categories. What this study contributes to the literature is a further distinction within formal reporting, highlighting that non-state institutions are actively used by GBV victims and, in some cases, potentially at even higher rates than state institutions. The predominance of social-circle reporting also points to a potential avenue for future data collection efforts through citizens themselves. It is clear that family members, neighbors and friends hold valuable information about GBV prevalence at the community or state level and further research is need to explore how they could help uncover unreported cases of violence.

Further, this study highlights the meaningful contribution of non-state institutions in capturing GBV experiences. Although most reporting occurred informally through social networks, non-state actors still appeared at a notable rate, even comparable to state reporting. Their presence indicates that they are an important resource for women, and that their data should be considered in policy discussions,

These results indicate that GBV victims reporting to state institutions are more likely to have higher educational attainment, live in urban areas, and have fewer children.

Additionally, the logistic regressions for state reporting ndicate that GBV victims reporting to state institutions are more likely to have higher educational attainment, live in urban areas, and have fewer children. Whereas GBV victims of older age, with lower education, who live rural residences, and have several young children were associated with higher odds of reporting to non-state channels. This pattern may reflect that urban areas are closer to state institutions, and that mobility is generally easier, while higher literacy facilitates interaction with many state resources. Surprisingly, the results also suggest that having more children is associated with lower likelihood of reporting to state institutions, which could be related to concerns about potential consequences or the practical challenges of leaving children to make a report. Further research is needed to explore these dynamics in greater detail.

The logistic regression results for non-state reporting paint a picture of women who are frequently overlooked by state institutions. This includes women who live in remote or rural areas far from police stations, making travel and access to formal reporting mechanisms difficult. It also encompasses women with limited literacy or education, who may feel intimidated by bureaucratic procedures or complex reporting forms, and those whose childcare responsibilities or household labor create time and logistical constraints that make engaging with state institutions challenging. The positionality of NGOs and community organizations not only is of a safer spaces for general disclosure, but also an accessible institutions for specific groups of women, women who are often missed by state mechanisms. Which suggests that NGOs, community organizations, religious institutions, and women's centers are critical intermediaries in both prevention and data collection of GBV. Their inclusion is essential when measuring GBV prevalence and designing policies that reflect women's diverse realities.

One limitation of this study is that the available dataset for self-initiated reporting does not allow for a direct regression comparison between state and non-state reporting, as these reporting options are not mutually exclusive. A more precise comparison would require data in which women are compelled to choose either state or non-state reporting. Alternatively, future surveys could employ experimental designs, such as vignettes, where respondents are asked to select one reporting pathway, enabling a clearer analysis of factors influencing their choice.

In the context of solicited reporting, the India case study indicates that more women disclosed GBV to the non-state survey (SNEHA) than to the state-led NFHS-4. This may suggest that women feel safer or more willing to report sensitive experiences when approached by non-state actors. SNEHA, a feminist NGO in Mumbai, trained interviewers extensively, provided referral systems for survivors, and used specific approaches such as question framing, all of which may have created a more supportive setting for disclosure. However, it is important to acknowledge the limitations. The two surveys differed in methodology, sampling, and target populations, which may partly explain the differences in prevalence estimates. A stronger test of this hypothesis would require parallel state and non-state surveys administered to the same population using the same questionnaire.

Another limitation is the inability to engage directly with NGOs, such as SNEHA, to explore their survey approaches and understand the experiences of GBV victims when reporting. Further qualitative research, with potential interviews should investigate these aspects to provide deeper insights.

Overall, these findings emphasize that non-state institutions occupy a special positionality in GBV reporting. They not only offer support and safe avenues for disclosure for women who might otherwise remain invisible to state systems, but they also capture data critical for understanding the true prevalence of GBV. Policymakers, researchers, and advocacy organizations should prioritize collaboration with NGOs, shelters, women's centers, and community-based organizations when designing interventions, collecting data, and interpreting prevalence statistics. Recognizing the complementary role of both state and non-state actors can significantly improve the reach, accuracy, and inclusivity of GBV prevention and response efforts.

7 Conclusion

This paper set out to answer the question: To whom do women report gender-based violence, and why? By proposing and empirically testing a two-dimensional typology of GBV disclosure, distinguishing self-initiated from solicited reporting, and state-affiliated from non-state-affiliated institutions, this study demonstrates that women's reporting behaviors cannot be adequately understood through the traditional "formal vs. informal" lens. Rather, the affiliation of the institution and the context of disclosure fundamentally shape whether GBV becomes visible, to whom, and under what conditions.

In the context of self-initiated reporting, the findings across four countries were clear: women overwhelmingly first disclose to their social circles before engaging with any institution. Beyond social networks, reporting rates to state institutions (police, courts, public health officials) and non-state institutions (NGOs, women's centers, community leaders, shelters) were comparable, with non-state reporting slightly higher in Kenya and Nigeria. Regression analyses further revealed that women who report to non-state institutions are more likely to have lower educational attainment, live in rural areas, and have more young children, women who are often structurally marginalized

by state systems. These findings suggest that non-state institutions serve as a critical safety net, reaching survivors who might otherwise remain invisible.

In the context of solicited reporting, the India case study reinforces this dynamic. Women disclosed experiences of GBV at higher rates in NGO-led surveys (SNEHA) than in state-administered national surveys (NFHS-4), highlighting how the nature of the actor requesting disclosure profoundly affects what is revealed. While methodological differences between the surveys limit causal claims, the patterns suggest that non-state actors may provide a more trusted and survivor-friendly environment, even in research contexts designed to ensure confidentiality.

Together, these findings underscore a central insight: women's NGOs, shelters, community organizations, and local support networks hold a unique positionality in the landscape of GBV reporting and data collection. Excluding their perspectives from national statistics and policy discussions risks systematically underestimating GBV prevalence and overlooking groups of survivors.

Ultimately, this study calls for a reframing of how GBV data is conceptualized, collected, and interpreted. Understanding women's reporting decisions is essential not only for accurately measuring GBV but also for designing effective responses. Recognizing, valuing, and integrating the work of women's organizations is therefore not optional, it is necessary for building a GBV policies that reflects the realities of women's lives and they futures they hold

References

Acker, J. (1990). Hierarchies, jobs, bodies: A theory of gendered organizations. *Gender & society*, *4*(2), 139–158.

Akurugu, C. A., Nyuur, R. B., & Dery, I. (2023). Non-governmental organizations' approaches to women's empowerment amid the covid-19 pandemic: Towards decolonizing development praxis in northern ghana. *World Development Sustainability*, *3*, 100079.

Ali, F. A. (2017). Women's agency and violence against women: The case of the coalition on violence against women in kenya. *African conflict and peacebuilding review*, *7*(1), 51–65.

Ansara, D. L., & Hindin, M. J. (2010). Formal and informal help-seeking associated with women's and men's experiences of intimate partner violence in canada. *Social science & medicine*, *70*(7), 1011–1018.

Borenstein, M., Hedges, L. V., Higgins, J. P., & Rothstein, H. R. (2010). A basic introduction to fixed-effect and random-effects models for meta-analysis. *Research synthesis methods*, *1*(2), 97–111.

Bumiller, K. (2008). *In an abusive state: How neoliberalism appropriated the feminist movement against sexual violence*. Duke University Press.

Cattaneo, L. B., & DeLoveh, H. L. (2008). The role of socioeconomic status in helpseeking from hotlines, shelters, and police among a national sample of women experiencing intimate partner violence. *American Journal of Orthopsychiatry*, *78*(4), 413–422.

Daruwalla, N., Kanougiya, S., Gupta, A., Gram, L., & Osrin, D. (2020). Prevalence of domestic violence against women in informal settlements in mumbai, india: A cross-sectional survey. *BMJ open*, *10*(12), e042444.

Edition, T. (2006). Principles of epidemiology. *Center for Disease Control*.

Ellsberg, M., Heise, L., Organization, W. H., et al. (2005). Researching violence against women: A practical guide for researchers and activists. *World Health Organisation*.

for Economic Co-operation, O., & Development. (2018). *Development co-operation report 2018: Joining forces to leave no one behind*. OECD Publishing.

Graaff, K. (2021). The implications of a narrow understanding of gender-based violence. *Feminist Encounters*, *5*(1), 1–12.

Gracia, E. (2004). Unreported cases of domestic violence against women: Towards an epidemiology of social silence, tolerance, and inhibition.

Hall, H. I., Correa, A., Yoon, P. W., Braden, C. R., for Disease Control, C., Prevention, et al. (2012). Lexicon, definitions, and conceptual framework for public health surveillance. *MMWR Surveill Summ*, *61*(Suppl), 10–4.

Helbing, D. (2014). The world after big data: What the digital revolution means for us. *Available at SSRN 2438957*.

Kumari, U. (2023). Exploring the role of ngos in addressing gender-based violence against women. *Journal of Advanced Research in Women's Studies*, *1*(1), 38–52.

LaValley, M. P. (2008). Logistic regression. *Circulation*, *117*(18), 2395–2399.

McQueen, K., Murphy-Oikonen, J., Miller, A., & Chambers, L. (2021). Sexual assault: Women's voices on the health impacts of not being believed by police. *BMC women's health*, *21*(1), 217.

Measure, D., et al. (2012). Demographic and health surveys.

Montesanti, S. R. (2015). The role of structural and interpersonal violence in the lives of women: A conceptual shift in prevention of gender-based violence. *BMC women's health*, *15*(1), 93.

Palermo, T., Bleck, J., & Peterman, A. (2014). Tip of the iceberg: Reporting and gender-based violence in developing countries. *American journal of epidemiology*, *179*(5), 602–612.

Richards, N., Pandolfelli, L., Bouziani, B., Ofosu-Baadu, B., & Carter, K. (2022). The role of administrative data in gender statistics: Supporting inclusive development for women and girls. *Journal of International Development*, *34*(2), 349–378.

Schubotz, D., McAlister, S., & Neill, G. (2025). Collecting research evidence to inform programming to tackle violence against young women and girls: The importance of diverse data for policy making. *Youth*, *5*(1), 7.

Stöckl, H., & Sorenson, S. B. (2024). Violence against women as a global public health issue. *Annual review of public health*, *45*(1), 277–294.

Teutsch, S. M., & Churchill, R. E. (2000). *Principles and practice of public health surveillance*. Oxford University Press, USA.

Umubyeyi, A., Mogren, I., Ntaganira, J., & Krantz, G. (2014). Women are considerably more exposed to intimate partner violence than men in rwanda: Results from a population-based, cross-sectional study. *BMC women's health*, *14*(1), 99.

United Nations General Assembly. (1993). Declaration on the elimination of violence against women, a/res/48/104 [Proclaimed by General Assembly resolution 48/104]. https://www.un.org/en/genocideprevention/documents/atrocity-crimes/Doc.21_declaration%20elimination%20vaw.pdf

Walby, S., Towers, J., Balderston, S., Corradi, C., Francis, B., Heiskanen, M., Helweg-Larsen, K., Mergaert, L., Olive, P., Palmer, E., et al. (2017). *The concept and measurement of violence against women and men*. Policy Press.

Willetts, P. (2002). What is a non-governmental organization. *Conventions, treaties and other responses to global issues*, *2*(11), 229–248.

Wilson, K., Ung Loh, J., & Purewal, N. (2018). Gender, violence and the neoliberal state in india. *Feminist Review*, *119*(1), 1–6.

A Appendix

A.1 Graphs

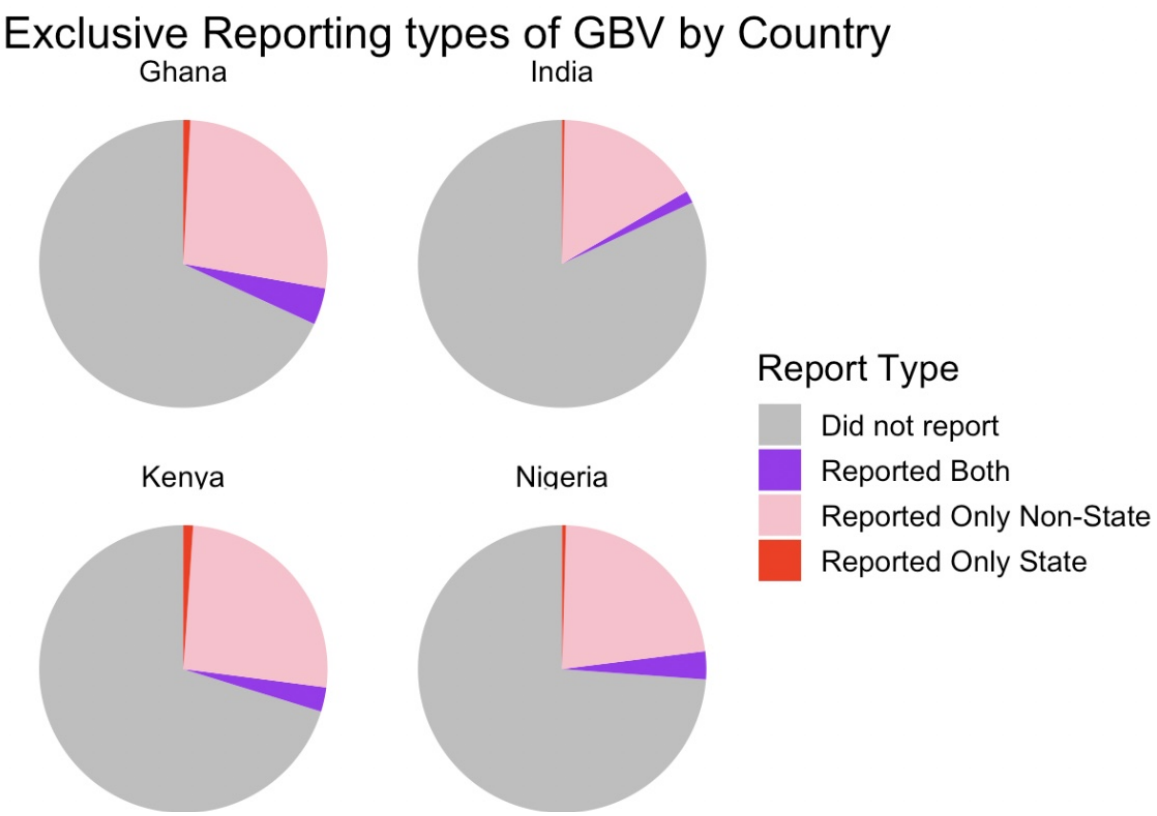


Figure 3: Pie charts displaying the reporting to state exclusively, non-state exclusively and reporting to both

A.2 Rcode used for analysis

```
```{r}
library(haven)
library(tidyverse)
library(dplyr)
```
```

```
library(data.table)
library(haven)
library(xtable)
library(flextable)
library(officer)
library(tidyr)
library(knitr)
library(kableExtra)

```

```{r, eval = FALSE}
#Importing data on Ghana 2008:
ghana_dhs <- read_dta("GHIR5ADT/GHIR5AFL.DTA")
head(ghana_dhs)
nrow(ghana_dhs) #Has 4916 observations
dim(ghana_dhs)

#Importing data on India 2005-2006.
india_dhs <- read_dta("IAIR52DT/IAIR52FL.dta")
head(india_dhs)
nrow(india_dhs)
dim(india_dhs)

#Importing data on Nigeria 2008
nigeria_dhs <- read_dta("NGIR53DT/NGIR53FL.DTA")
head(nigeria_dhs)
nrow(nigeria_dhs)
dim(nigeria_dhs)

#Variables needed for the study:
```

```r
cols_needed <- c(
  #Identifiable variables
  "caseid","v012","v020","v155","v156","v106","v107","v133",
  "v149","v190","v191","v025","v024","v102","v137", "v044",

  #any emotional abuse my husband
  "d103b",
  #any Physical violence (less, and severe)
  "d106", "d107", "d130a","d118y", "d115y",

  #any sexual violence
  "d108", "d130b", "d124", "d125",

  #Reporting and disclosing GBV
    "d128","d119y"
)

#Importing Ghana DHS GBV Data

#Importing Ghana 2008
ghana_2008 <- read_dta("GHIR5ADT/GHIR5AFL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Ghana", Year = 2008)

#Not importing Ghana 2014 because GBV module

#Importing Ghana 2022
ghana_2022 <- read_dta("GHIR8CDT-2022/GHIR8CFL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Ghana") %>%
  mutate(Year = 2022)

#Importing India DHS GBV Data
```

```r

#Importing India data 2005-2006
india_2005 <- read_dta("IAIR52DT/IAIR52FL.dta")%>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "India", Year = 2005)


#Importing India 2015
# They are both too big, had to use fread
india_2015 <- read_dta(
  "IAIR74DT-2015-2016/IAIR74FL.DTA",
  col_select = c(any_of(cols_needed), starts_with("d119"))
) %>%
  mutate(Country = "India", Year = 2015)

#Importing India 2020
india_2020 <- read_dta(
  "IAIR7EDT-2019-2021/IAIR7EFL.DTA",
  col_select = c(any_of(cols_needed), starts_with("d119"))
) %>%
  mutate(Country = "India", Year = 2020)

#Importing  Nigeria DHS GBV Data
#Importing Nigeria 2008
nigeria_2008 <- read_dta("NGIR53DT/NGIR53FL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Nigeria", Year = 2008)

#Importing Nigeria 2013
nigeria_2013 <- read_dta("NGIR6ADT-2013/NGIR6AFL.DTA")%>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Nigeria") %>%
  mutate(Year = 2013)
```

```r

#Importing Nigeria 2018

nigeria_2018 <- read_dta("NGIR7BDT-2018/NGIR7BFL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Nigeria") %>%
  mutate(Year = 2018)

#Importing Nigeria 2024

nigeria_2024 <- read_dta("NGIR8BDT-2024/NGIR8BFL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Nigeria") %>%
  mutate(Year = 2024)

#Importing Kenya DHS GBV Data

#Importing Kenya 2008
kenya_2008 <- read_dta("KEIR52DT-2008/KEIR52FL.DTA") %>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Kenya") %>%
  mutate(Year = 2008)

#Importing Kenya 2014
kenya_2014 <- read_dta("KEIR72DT-2014/KEIR72FL.DTA")%>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Kenya") %>%
  mutate(Year = 2014)

#Importing Kenya 2022
kenya_2022 <- read_dta("KEIR8CDT-2022/KEIR8CFL.DTA")%>%
  select(any_of(cols_needed), starts_with("d119")) %>%
  mutate(Country = "Kenya") %>%
```

```
  mutate(Year = 2022)


## Data Wrangling:

# Combining all surveys from different years all together (adding more years)
#Ghana:
allyears_ghana <- suppressWarnings(bind_rows(ghana_2008, ghana_2022))

allyears_ghana

#India

allyears_india <- suppressWarnings(bind_rows(india_2005, india_2015, india_2020))


#Nigeria
all_years_nigeria <- suppressWarnings(bind_rows(nigeria_2008, nigeria_2013, nigeria_2018, nigeria_2024))

#Kenya

all_years_kenya <- suppressWarnings(bind_rows(kenya_2008, kenya_2014, kenya_2022))



#Combining all datasets for ALL years survey for all countries.
combined_dhs<- suppressWarnings(bind_rows(allyears_ghana, allyears_india, all_years_nigeria, all_years_kenya))  %>%
```

```
  filter(v044 == 1) #Filtering for women age 15-49 selected and
     interviewed for the domestic violence module

head(combined_dhs_2)
nrow(combined_dhs_2)
ncol(combined_dhs_2)
```

The dataset has 362588 observations of women between 15-49 who were interviewed on the GBV module from 4 countries (India, Kenya, Ghana, Nigeria) and 65 variables of interest.

```{r}
#Wrangling new dataset:


#Creating new variable names:
combined_dhs_2<- combined_dhs%>%
  mutate(
  reported_to_family =
    as.integer(rowSums(across(d119a:d119r), na.rm = TRUE) > 0),

  reported_to_friend =
    as.integer(rowSums(across(c(d119s, d119t, d119xd)), na.rm = TRUE) > 0)
       ,

  reported_to_other =
    as.integer(rowSums(across(c(d119xa, d119x)), na.rm = TRUE) > 0),

  reported_to_cs_organisation =
```

```
    as.integer(rowSums(across(c(d119xc, d119xi, d119xj, d119xk)), na.rm =
        TRUE) > 0)
) %>%
  mutate(across(
    c( d119u, d119xb, d119xf, d119w, d119xe, d119xg, d119xh, d119v),
    ~ as.integer(replace_na(.x, 0) > 0)
  )) %>%
  mutate(across(c(d119y, d118y), ~ as.integer(replace_na(.x, 1) == 1)))
     %>%
  rename(
    "caseid" = caseid,
    "current age - respondent" = v012,
    "ever-married sample" = v020,
    "literacy" = v155,
    "ever participated in a literacy program outside of primary" = v156,
    "highest educational level" = v106,
    "highest year of education" = v107,
    "education in single years" = v133,
    "educational attainment" = v149,
    "wealth index" = v190,
    "wealth index factor score (5 decimals)" = v191,
    "type of place of residence v025" = v025,
    "type of place of residence v024" = v024,
    "type of place of residence v102" = v102,
    "number of children 5 and under in household (de jure)" = v137,
    "Ever been threatened with harm by husband/partner" = d103b,
    "Experienced any less severe violence" = d106,
    "Experienced any severe violence" = d107,
    "Previous husband: ever hit, slap, kick, or physically hurt respondent
        " = d130a,
    "Respondent was not hurt by anyone during a pregnancy" = d118y,
    "No person other than husband/partner ever physically hurt respondent"
         = d115y,
```

```
    "Experienced any sexual violence" = d108,
    "Previous husband/partner: physically forced to have sex or to perform
        sexual acts" = d130b,
    "Ever forced to have sex by anyone other than husband/partner in last
       12 months" = d124,
    "Ever forced to perform unwanted sexual acts" = d125,
    "Have ever told anyone else about violence" = d128,
    "Respondent did not seek help from anyone" = d119y,
    "reported_to_neighbour" = d119u,
    "reported_to_Social_service_organisation" = d119xb,
    "reported_to_Religious_leader" = d119xf,
    "reported_to_Employer" = d119w,
    "reported_to_Police" = d119xe,
    "reported_to_Criminal_Justice_System" = d119xg,
    "reported_to_Public_health_official" = d119xh,
    "reported_to_Public_education" = d119v,
    "Country" = Country,
    "Year" = Year
  ) %>%
  select(-(starts_with("d119")))  %>%
  #Creating GBV variable
  mutate(
    experienced_GBV = if_else(
      # Condition: either any GBV variable >= 1
      rowSums(across(c(
        `Ever been threatened with harm by husband/partner`,
        `Experienced any less severe violence`,
        `Experienced any severe violence`,
        `Experienced any sexual violence`,
        `Previous husband: ever hit, slap, kick, or physically hurt
           respondent`,
        `Previous husband/partner: physically forced to have sex or to
           perform sexual acts`,
```

```
        `Ever forced to have sex by anyone other than husband/partner in
           last 12 months`,
        `Ever forced to perform unwanted sexual acts`
      ), ~ .x >= 1), na.rm = TRUE) > 0
      # OR if either of the "no violence" indicators is 0
      | replace_na(`No person other than husband/partner ever physically
         hurt respondent`, 1) == 0
      | replace_na(`Respondent was not hurt by anyone during a pregnancy`,
          1) == 0,
      1,  # value if TRUE
      0   # value if FALSE
    )) %>%
  mutate(`education in single years` = if_else(`education in single years`
      == 99,NA, `education in single years`)) %>%
  mutate(`ever participated in a literacy program outside of primary` = if
     _else(`ever participated in a literacy program outside of primary` ==
      9,NA, `ever participated in a literacy program outside of primary`))
      %>%
  mutate(`educational attainment` = if_else(`educational attainment` == 9,
      NA, `education in single years`))

  library(scales)

# Summarize data by Country and Reported_GBV
bar_graph_2 <- combined_dhs_2 %>%
  filter(experienced_GBV == 1) %>%
  filter(!is.na(`Respondent did not seek help from anyone`)) %>%
  mutate(
    Report_GBV = `Respondent did not seek help from anyone` == 0,
    Reported_GBV = if_else(Report_GBV, "Reported", "Did not report")
  ) %>%
  group_by(Country, Reported_GBV) %>%
  summarise(n = n(), .groups = "drop") %>%
```

```r
  group_by(Country) %>%
  mutate(pct = n / sum(n))  # calculate proportion for stacked bar

# Plot 100% stacked bar chart
ggplot(bar_graph_2, aes(x = Country, y = pct, fill = Reported_GBV)) +
  geom_col() +
  geom_text(data = filter(bar_graph_2, Reported_GBV == "Reported"),
            aes(label = paste0(round(pct*100), "%")),
            position = position_stack(vjust = 0.5), color = "white") +
  scale_fill_manual(values = c("Reported" = "red", "Did not report" = "
      darkred")) +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Percentage of Self-Initiated GBV Reporting by Country, 2005
         2024  ",
    x = "Country",
    y = "Experienced GBV",
    fill = "Reported GBV"
  ) +
  theme_minimal(base_size = 14)

#Creating variables for different reporting modes:

Data_with_3_reporting_modes <- combined_dhs_2 %>%
  mutate(state_reporting = if_else(
      rowSums(across(c(reported_to_Police,reported_to_Criminal_Justice_
          System, reported_to_Public_health_official, reported_to_Public_
          education),
                     ~ .x >= 1), na.rm = TRUE) > 0,
      1, 0
    )) %>%
  mutate(non_state_reporting = if_else(
```

```
304       rowSums(across(c(reported_to_Social_service_organisation, reported_
              to_Religious_leader, reported_to_cs_organisation, reported_to_
              Employer),
305                      ~ .x >= 1), na.rm = TRUE) > 0,
306       1, 0
307     )) %>%
308   mutate(social_circle_reporting = if_else(
309       rowSums(across(c(reported_to_neighbour, reported_to_family, reported
              _to_friend, reported_to_other),
310                      ~ .x >= 1), na.rm = TRUE) > 0,
311       1, 0
312     ))
313
314 Data_with_3_reporting_modes
315
316 #Summary stats of 3 modes of reporting:
317
318 Reporting_proportion<- Data_with_3_reporting_modes %>%
319    filter(experienced_GBV == 1) %>%
320   group_by(Country) %>%
321   summarise(
322     total_experienced = n(),
323     total_reported = sum(`Respondent did not seek help from anyone` == 0,
          na.rm = TRUE),
324     pct_reported = 100 * (sum(`Respondent did not seek help from anyone`
          == 0, na.rm = TRUE)/total_experienced),
325     total_reported_state = sum(state_reporting == 1, na.rm = TRUE),
326     pct_reported_state = 100 * (sum(state_reporting == 1, na.rm = TRUE)/
          total_experienced), #How many of those reported went to state
327     total_reported_non_state = sum(non_state_reporting == 1, na.rm = TRUE)
          , #How many of those reported went to non-state
328     pct_reported_non_state = 100 * (sum(non_state_reporting == 1, na.rm =
          TRUE)/ total_experienced),
```

```
    total_reported_social_circle = sum(social_circle_reporting == 1, na.rm
        = TRUE),
    pct_reported_social_circle = 100 * (sum(social_circle_reporting == 1,
       na.rm = TRUE)/ total_experienced)
  )

Reporting_proportion


library(tidyverse)

# Reshape data to long format for ggplot
Reporting_long <- Reporting_proportion %>%
  select(Country, pct_reported_state, pct_reported_non_state, pct_reported
     _social_circle) %>%
  pivot_longer(
    cols = starts_with("pct_reported"),
    names_to = "Reporting_Type",
    values_to = "Percentage"
  ) %>%
   mutate(
    Reporting_Type = case_when(
      Reporting_Type == "pct_reported_state" ~ "State",
      Reporting_Type == "pct_reported_non_state" ~ "Non-State",
      Reporting_Type == "pct_reported_social_circle" ~ "Social Circle"
    ),
    # Relevel factor to control stacking order
    Reporting_Type = factor(Reporting_Type, levels = c("Social Circle", "
       Non-State", "State"))
  )

# Plot
```

```r
ggplot(Reporting_long, aes(x = Country, y = Percentage, fill = Reporting_Type)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = paste0(round(Percentage, 1), "%")),
            position = position_stack(vjust = 0.5), color = "white", size = 3) +
  scale_fill_manual(values = c("State" = "#CC2F4D",          # dark pink
                               "Non-State" = "#E85A8C",     # medium pink
                               "Social Circle" = "#F4A6C1")) + # light pink
  labs(
    y = "Reported GBV (%)",
    x = "Country",
    fill = "Reporting Type",
    title = "Proportion of GBV Reporting by Country and Type"
  ) +
  theme_minimal() +
  theme(
    text = element_text(size = 12),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )


#A logistic regression for state reporting fiex effects on country
library(fixest)

state_reporting_model <- feglm(
  state_reporting ~ experienced_GBV +
                    `Respondent did not seek help from anyone`  +
                     `current age - respondent` +
                     `educational attainment` +
                     `wealth index factor score (5 decimals)` +
                      `type of place of residence v025` +
```

```
                    `number of children 5 and under in household (de jure
                       )`,
  data = Data_with_2_reporting_modes,
  family = binomial(link = "logit"),
  fixef = "Country"   # fixed effect for country
)
summary(state_reporting_model)

non_state_reporting_model <- feglm(
  non_state_reporting ~ experienced_GBV +
                    `Respondent did not seek help from anyone`  +
                     `current age - respondent` +
                     `educational attainment` +
                     `wealth index factor score (5 decimals)` +
                      `type of place of residence v025` +
                     `number of children 5 and under in household (de jure
                       )`,
  data = Data_with_2_reporting_modes,
  family = binomial(link = "logit"),
  fixef = "Country"   # fixed effect for country
)
summary(non_state_reporting_model)
```
```
```